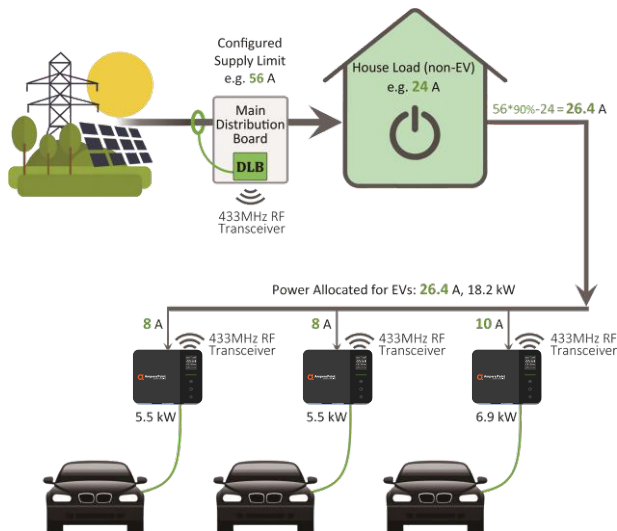


Quick User Manual

DLB Controller for Mode 3 Wall Box Charger

Model Number: DLB-A1

Applies to the household/small commercial DLB controller. This concise manual summarizes functions, safety, installation, pairing and operation, indicators, and typical current allocation examples for 1–3 wall box chargers. Appendices include a DIP quick lookup and 433 MHz antenna recommendations & coverage test method.



1. Scope & Safety

- Install and set up only by qualified electricians.
- Isolate power before wiring; lock-out/tag-out where applicable.
- **Current Transformer (CT)** orientation must be correct; never energize with CT secondary open-circuited.
- Set the **Configured Supply Limit (CSL)** according to incoming cable size, breaker rating, and local codes.
- Install inside a dry, dust-free, ventilated distribution board; this device is not weatherproof.
- Keep the wireless antenna away from strong EMI sources; ensure proper strain relief on cables.

2. Description & Operating Principle

The DLB monitors the household's total current (single/three-phase) and allocates optimal charging current to 1–3 wall boxes in real time, so that the site limit is never exceeded.

- **Dynamic Load Balancing:** per-EV optimization to use every available amp without exceeding the limit.
- **Safety Buffer:** default usable ceiling is 90% of the **Configured Supply Limit** (10% buffer).
- **Communications:** 433 MHz wireless (SMA antenna) or RS485; target current is delivered to wall box in real time.
- **Fairness & Constraints:** each wall box is capped by its own requested current; any branch < minimum sustaining current is paused.

3. Parts & I/O (Refer to Photo)

- **Antenna** (Gold-plated SMA): 433 MHz wireless.
- **Power input:** AC 100–240 V, 50/60 Hz, L / N / PE.
- **CT inputs:** L1 / L2 / L3 Current Transformers.
- **Pairing button:** **Hold 5 s** to enter pairing mode (**clears all existing links**); **press once** to finish early (otherwise times out after ~2 min).
- **Link LEDs:** link status for up to three wall boxes.
- **8-way DIP switch:** sets **Configured Supply Limit** (1–255 A).

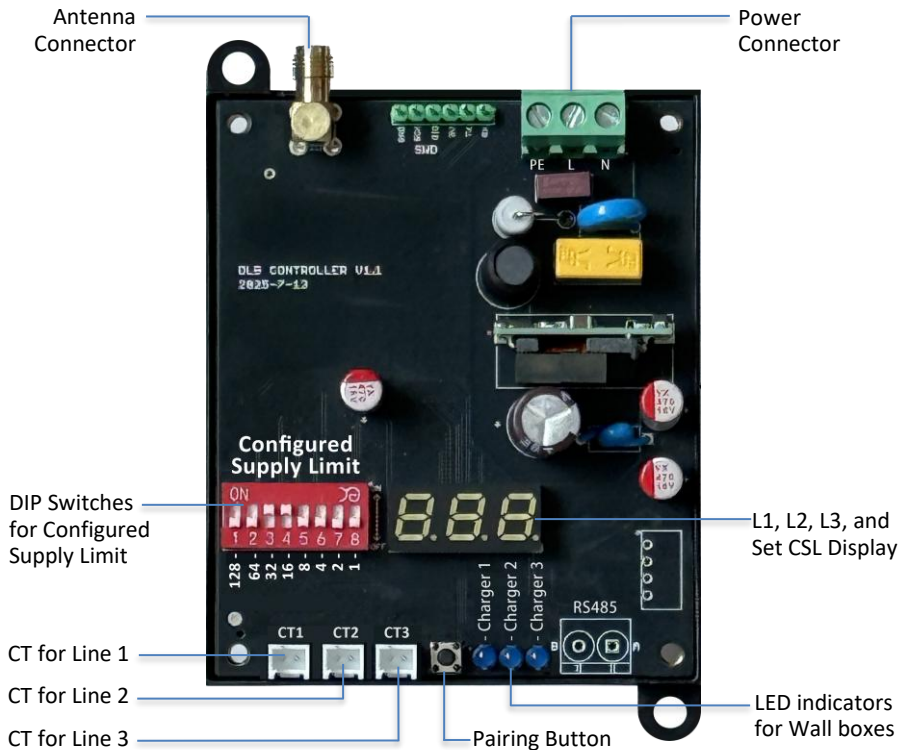


Figure: DLB controller — external connectors and indicators.

4. Installation

- Mount inside the main distribution board for short CT runs and convenient power/RS485 wiring.
- **CTs**: Clamp on the incoming L1/L2/L3, upstream of all branch breakers (not on branches). Arrow toward load; on three-phase, monitor all phases and limit by the heaviest phase.
- **433 MHz antenna** (default): keep vertical, ideally outside metal enclosures. Where site conditions—e.g., walls or large metal structures—impede wireless connectivity, select RS485.
- **RS485**: use either 433 MHz or RS485 — not both. Switching to RS485 requires opening the wall box and moving the internal selector to “RS485” before installation. Wire A/B correctly.

5. Settings

5.1 Configured Supply Limit (DIP 1–255 A)

Use the 8-way DIP (weights 128/64/32/16/8/4/2/1 A). Toggle ON to sum to the desired amperage. The DLB applies 90% of this value as the usable ceiling (10% safety buffer).

Examples: 75 A = 64+8+2+1; 62 A = 32+16+8+4+2; 50 A = 32+16+2.

5.2 Minimum Sustaining Current

Recommended 6 A. If a branch would receive <6 A, that wall box is paused until headroom returns.

6. Pairing & Operation

6.1 Pairing

- **Warning**: entering pairing mode clears all existing links; please re-pair ALL wall boxes (1–3) simultaneously.
- **DLB**: hold the Pairing Button for 5 s until all three link LEDs flash; the pairing mode times out after 2 min.
- **Wall box**: while not charging, press and hold A + C for 5 s → Settings → DLB Pairing → On → Confirm → Home. Blue DLB icon (flashing) indicates pairing mode.
- **Success**: corresponding DLB link LED turns solid; wall box DLB icon becomes green. Press the Pairing Button once to finish pairing early.

6.2 Operation

- **Full Current Charging** (Green DLB icon and white current color): sufficient headroom; each wall box charges at its requested current.
- **Reduced Current Charging** (Orange DLB icon and current color): headroom drops; DLB reduces and optimizes currents while keeping each branch ≥ 6 A where possible. Otherwise, that branch is paused.
- **Paused** (Red DLB icon and current color): This wall box is paused due to severe current shortage. Recovery is automatic when headroom returns.

7. Indicators

Mode	DLB LED	Wall Box Charger	
		DLB icon	Real-time Current
Not Paired/offline	Off	Grey	White
Pairing	Flashing	Flashing blue	White
Paired/online	Solid On	Green	White
Full Current Charging	Slow Flashing	Green	White
Reduced Current Charging	Slow Flashing	Orange	Orange
Charging Paused	Quick Flashing	Red	Red

8. Typical Allocation Examples

Assumptions:

- **Configured Supply Limit** set to 74 A → DLB usable limit = 66 A (90%);
- Usable Total Current = 66A - CT-measured current of house load(non-EV)
- each wall box requests up to 32 A;
- minimum sustaining current = 6 A; integer-amp allocations.

8.1 One Wall Box

Scenario	Usable Total (A)	WB1 (A)	DLB Icon	Current Color	Notes
Full Current	56	32	Green	White	Capped by wall box max 32 A.
Reduced Current	21.3	21	Orange	Orange	Limited by household load.
Paused	0–5	0	Red	Red	Below 6 A → paused.

8.2 Two Wall Boxes

Scenario	Usable Total (A)	WB1 (A)	WB2 (A)	DLB Icons	Current Colors	Notes
Full Current	56	32	16	Green / Green	White / White	All requests satisfied; no derating.
Reduced Current	26.5	17	9	Orange / Orange	Orange / Orange	Proportional reduction.
Critical	9.6	9	0	Orange / Red	Orange / Red	WB2 <6 A → pause WB2.

8.3 Three Wall Boxes

Scenario	Usable Total (A)	WB1 (A)	WB2 (A)	WB3 (A)	DLB Icons	Current Colors	Notes
Full Current	64	32	16	16	Green / Green / Green	White / White / White	All requests satisfied; no derating.
Reduced Current	35	18	9	8	Orange / Orange / Orange	Orange / Orange / Orange	Proportional reduction.
Critical	9.6	9	0	0	Orange / Orange / Red	Orange / Orange / Red	WB3 <6 A → pause WB3.

9. Key Specifications

Model Number	DLB-A1
Power Supply	AC 100–240 V, 50/60 Hz
Managed Wall Boxes	1-3
Communications	433 MHz (SMA) or RS485 (one link type at a time)
433 MHz Range	20–30 m indoor (typ.); 80–100 m line-of-sight (open field). Range is site-dependent (walls, reinforced concrete, metalwork, RF interference).
CT Inputs	L1 / L2 / L3, 0-100 A (2000:1), Accuracy class: 0.5
Configured Supply Limit	1–100 A, 8-way DIP switch, expandable to 150 / 200 / 250 A (with appropriate CT).
Usable Limit	90% × Configured Supply Limit minus the highest CT-measured phase current
Min Sustaining Current	6 A for each wall box
Overcurrent Reaction (downshift)	< 1 s (typ.)
Operating Environment	–20 to +55 °C, ≤95% RH (non-condensing)

Appendix A —DIP Quick Lookup

DIP weights: 128 / 64 / 32 / 16 / 8 / 4 / 2 / 1 (A). Toggle to ON to sum to the target amperage. The device applies a 10% safety buffer to the **Configured Supply Limit**.

Common examples:

Configured Supply Limit (A)	ON Weights (A)	Binary (1=ON from 128→1)	DIP Switch Settings (⬜=OFF, ⬛=ON)
25	16+8+1	00011001	⬛⬛⬛⬛⬛⬛⬜⬜
30	16+8+4+2	00011110	⬛⬛⬛⬛⬛⬛⬜⬜
35	32+2+1	00100011	⬛⬛⬛⬛⬛⬛⬛⬜
42	32+8+2	00101010	⬛⬛⬛⬛⬛⬛⬛⬜
48	32+16	00110000	⬛⬛⬛⬛⬛⬛⬜⬜
50	32+16+2	00110010	⬛⬛⬛⬛⬛⬛⬛⬜
55	32+16+4+2+1	00110111	⬛⬛⬛⬛⬛⬛⬛⬜
62	32+16+8+4+2	00111110	⬛⬛⬛⬛⬛⬛⬛⬜
65	64+1	01000001	⬛⬜⬛⬛⬛⬛⬛⬛⬜
68	64+4	01000100	⬛⬜⬛⬛⬛⬛⬛⬛⬜
70	64+4+2	01000110	⬛⬜⬛⬛⬛⬛⬛⬛⬜
75	64+8+2+1	01001011	⬛⬜⬛⬛⬛⬛⬛⬛⬜
78	64+8+4+2	01001110	⬛⬜⬛⬛⬛⬛⬛⬛⬜
80	64+16	01010000	⬛⬜⬛⬛⬛⬛⬛⬜⬜
85	64+16+4+1	01010101	⬛⬜⬛⬛⬛⬛⬛⬛⬜
90	64+16+8+2	01011010	⬛⬜⬛⬛⬛⬛⬛⬛⬜
93	64+16+8+4+1	01011101	⬛⬜⬛⬛⬛⬛⬛⬛⬜
95	64+16+8+4+2+1	01011111	⬛⬜⬛⬛⬛⬛⬛⬛⬜
100	64+32+4	01100100	⬛⬜⬛⬛⬛⬛⬛⬜⬜

Appendix B — 433 MHz Antenna Installation Recommendations

- **Antenna type:** 433 MHz $\frac{1}{4}$ -wave whip with SMA and magnetic base.
- **Mounting:** Install the DLB inside the distribution board for short Power and CT runs. Route the antenna cable out of the metal enclosure and mag-mount the base on the exterior metal surface (top or side).
- **Orientation:** Keep the whip vertical. It's omnidirectional—no need to point it at the wall box.
- **Distance & path:** Keep the DLB–wall box distance as short as practical and aim for a clear path with no large metal objects in between.
- **Height & placement:** Prefer the top of the cabinet or another high, open spot; avoid corners, deep recesses, and spots shielded by beams/ducts.
- **Ground plane:** A magnetic metal surface under the base acts as a ground plane—good for performance. Keep the whip a few centimeters away from other metal masses or rails.
- **Separation:** Avoid coiling excess cable near power devices; keep several centimeters of clearance from large metal obstacles.
- **Cable routing:** Stay away from high-current cables/VFDs; if you must cross them, do it at right angles. Provide basic strain relief; avoid sharp bends.
- **Connectors:** Hand-tighten the SMA; check for a firm connection after positioning.
- **Environment:** If mounted where moisture/dust is possible, add basic weather shielding (indoor use only unless otherwise rated).
- **Regulatory:** Follow local ISM-band limits (transmit power/duty cycle).